

A temporal perspective of facilitation during forest recovery in Mediterranean old fields

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The recovery of Mediterranean forests after field abandonment is slow, mainly because of high seedling mortality

Plant facilitation (especially of nurse shrubs) play a critical role in forest succession by ameliorating the environmental constraints for tree recruitment

We investigated the interaction between the colonization of the nurse shrub *Retama sphaerocarpa* and the recruitment of two oak species with contrasting ecological strategies in old fields abandoned 30 years ago

20 years after field abandonment is the time needed by nurse shrubs to accelerate forest succession

7 years is the minimum age of the nurse shrubs for facilitating oak recruitment

2.2 % was the needed cover of nurse shrub canopy able to accelerate oak recruitment

47 mm of summer precipitation marked the water stress limit under which facilitation occurred

(Solution is at the bottom)

Try to date this *Retama sphaerocarpa* sample. It is funny to count growth rings!

No, I'm not a plant lover

Do you want to know more?

Yes, it seems exciting

Let's travel to deep Spain...

Yes, but I have not much time

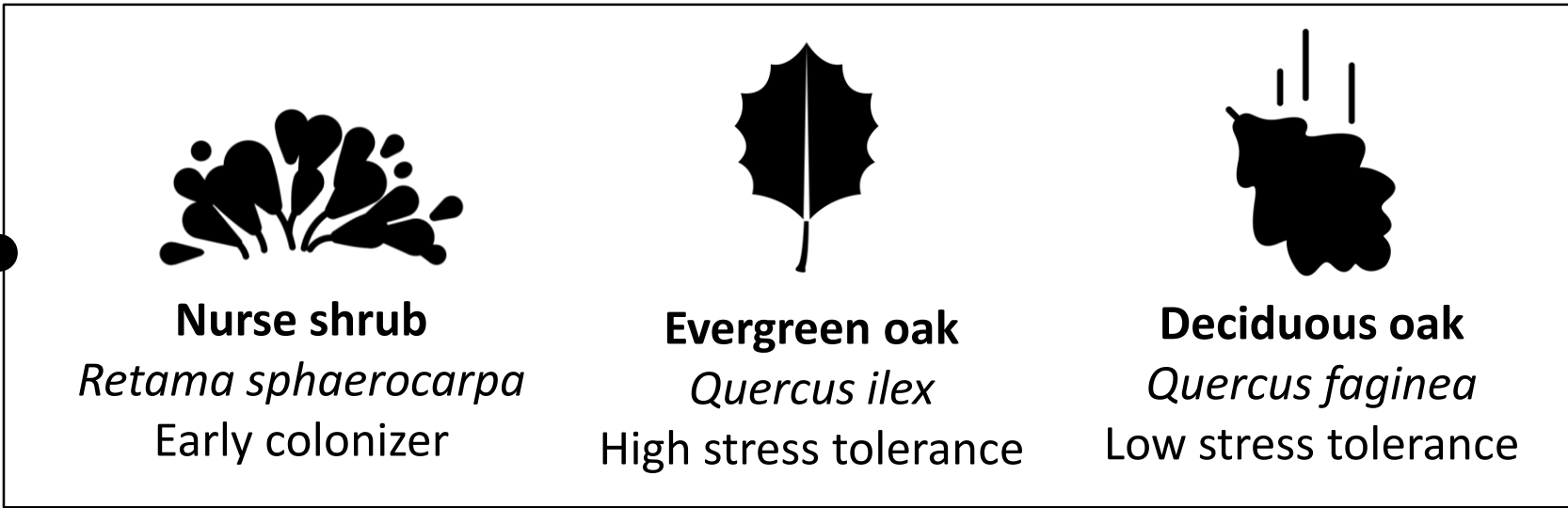
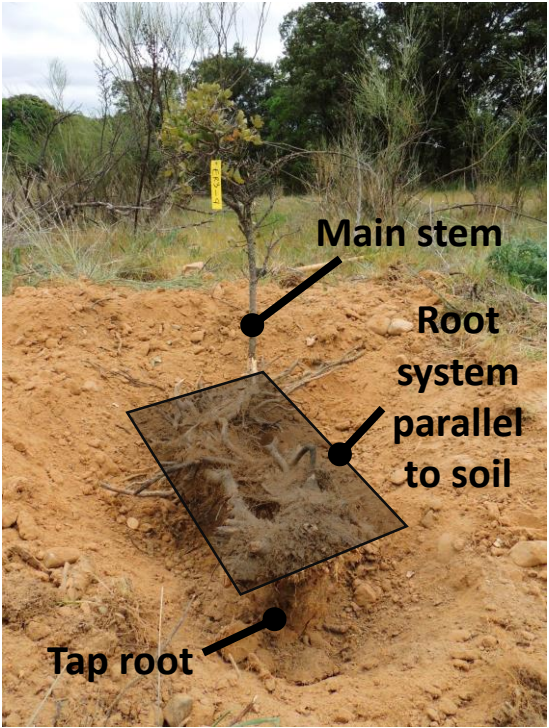
Read these key messages



1. What did we do there?

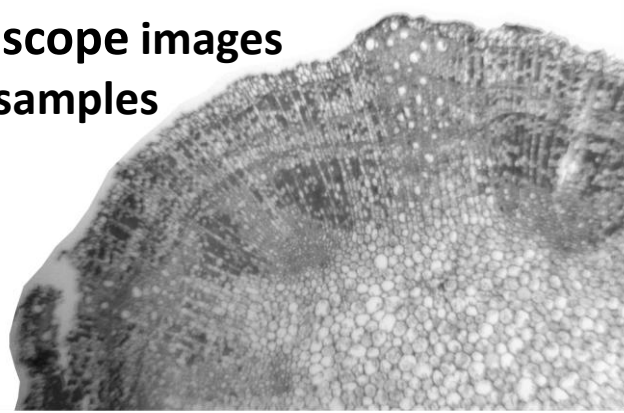
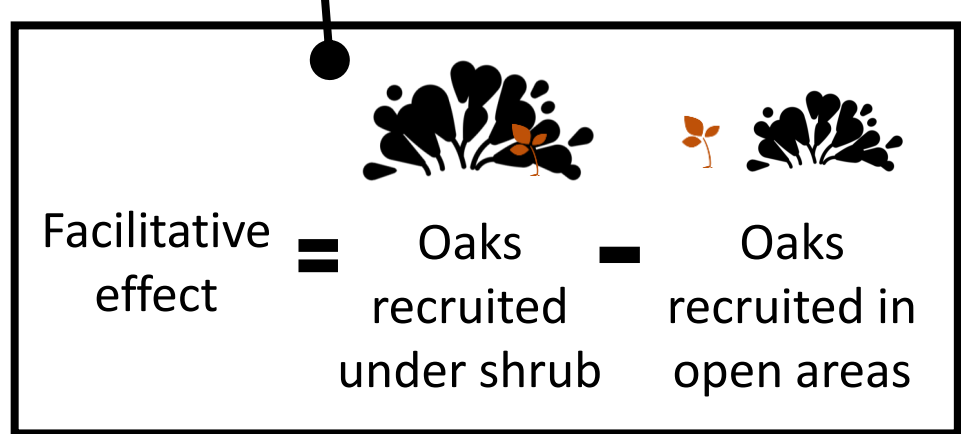
- We quantified the annual colonization of three species...

...by extracting the root collar of the shrubs and young oaks in the abandoned fields...



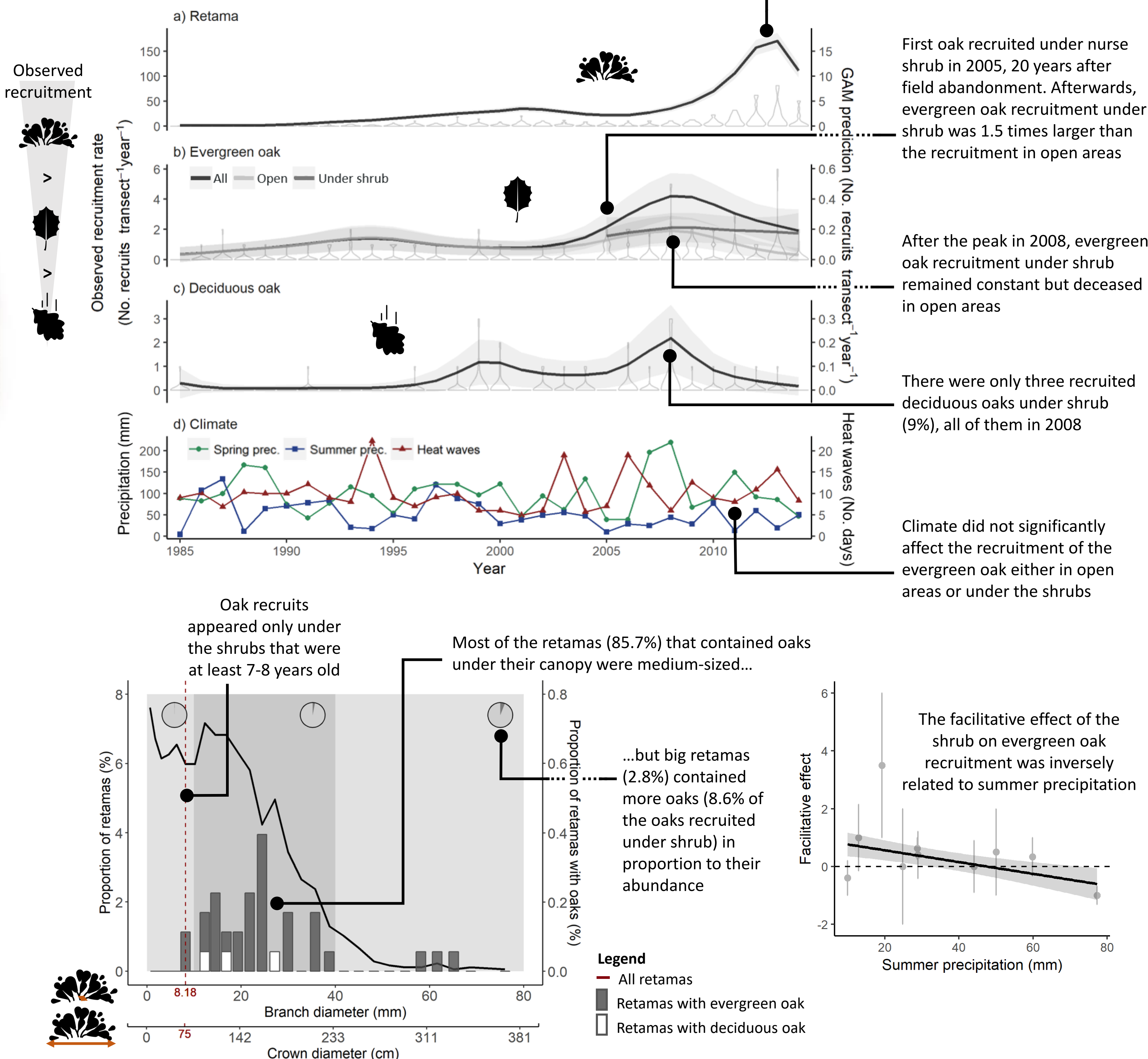
...and dating the samples using dendrochronology:

- We fitted generalised mixed models to the yearly recruitment and to the facilitative effect



2. What did we come up with?

The temporal trends in pulses of recruitment suggest distinct regeneration dynamics for the shrub and the two oak species



These results point to evergreen oak dominance in the secondary forest in the mid-term under more-arid climate scenarios

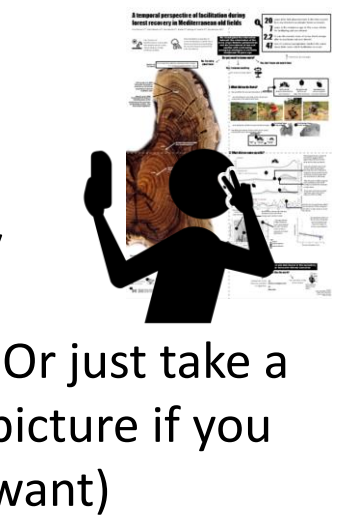
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